CYS 331 —Unit 5: Bluetooth Security

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Learning Outcomes

- Describe Bluetooth basics, network topologies, and the protocol family security model.
- Explain how pairing, authentication, and encryption establish trusted short-range communication.
- Discuss piconets, scatternets, baseband protection, and frequency hopping clearly.
- Evaluate Bluetooth threats and outline a basic security assessment process.

Basic Specifications and Use Cases

- Bluetooth is a short-range wireless technology designed for personal device connectivity.
- It supports audio, peripherals, sensing, control, and low-power data exchange.
- Core specifications define radio behavior, roles, services, and security features.
- Security understanding begins with knowing the protocol family and device capabilities.

Piconets

- A piconet links one central device with several participating peripheral devices.
- Communication timing and coordination depend on the central device control role.
- Piconets enable efficient short-range networking for personal and embedded devices.
- Security depends on trustworthy pairing, device identity, and session management.

Scatternets

- Scatternets connect multiple piconets through devices participating in more than one network.
- They expand coverage and flexibility, but increase coordination and security complexity.
- Routing across linked groups creates additional trust and availability concerns.
- Designers must manage identity, synchronization, and segmentation carefully in larger topologies.

Bluetooth Security Architecture

- Bluetooth security architecture includes pairing, key generation, authentication, authorization, and encryption.
- Trust is not automatic; it is established through specific security procedures.
- Security modes and service permissions determine what connected devices may do.
- Misconfiguration can expose services even when devices appear properly paired.

Baseband, Link Layer, and Frequency Hopping

- Baseband and link-layer controls coordinate timing, addressing, retransmission, and error handling.
- Frequency hopping reduces interference by changing channels in a structured sequence.
- These mechanisms improve robustness, but do not replace authentication and encryption.
- Layered protection is needed because radio resilience alone cannot stop attackers.

Security Manager and Pairing

- The security manager coordinates pairing, bonding, key storage, and trust decisions.
- Pairing methods vary in usability, assurance, and resistance to impersonation.
- Weak pairing choices create long-term exposure because trust may persist.
- Users should verify devices carefully before accepting new trust relationships.

Authentication and Encryption

- Authentication checks whether communicating devices are genuinely who they claim.
- Encryption protects session data from nearby interception during wireless transmission.
- Strong keys and modern algorithms greatly improve confidentiality and integrity.
- Security fails when devices skip verification or fall back unnecessarily.

Threats and Security Assessment

- Threats include eavesdropping, impersonation, unauthorized pairing, tracking, and service abuse.
- Security assessment reviews discoverability, pairing behavior, services, logs, and update status.
- Analysts test both technical controls and human interaction weaknesses.
- Practical defense combines secure configuration, patching, policy, and user awareness.

Unit 5 Summary

- Bluetooth security depends on topology, trust establishment, and careful device configuration.
- Piconets and scatternets change coordination needs and expand security complexity.
- Pairing, authentication, and encryption work together to protect short-range communication.
- Assessment should test services, trust decisions, updates, and human behavior.